40 Particle accelerators

The first subatomic particles to be discovered were electrons. These have very little mass (about 1/2000 the mass of the lightest atom) and are negatively charged, which makes them easy to accelerate up to high speeds. For instance in a domestic television set the electrons may be accelerated using a potential difference of 11 000 V, which gives them a speed of approximately 62 000 000 ms^{-1}. Since this is a significant fraction of the speed of light, relativistic mechanics must be used for accurate calculations.

To accelerate charged particles to higher energies it is convenient to make them move in circular paths, so that they can be accelerated repeatedly. This is done by making them travel in a magnetic field, and was first put into practice in 1930 by E. Lawrence. His 'cyclotron' had a diameter of 13 cm and accelerated protons to an energy of 80 000 eV. The principle of its construction is shown in the diagrams, figure 1 and figure 2.

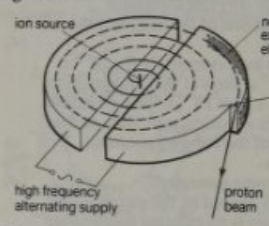

Figure 1

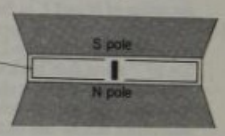

Figure 2

At the centre was a source of protons, which would be accelerated to whichever 'D' happened to be negative, then travel in a semicircle to arrive back at the gap when the *other* 'D' was negative. The proton would therefore be accelerated again, and so on, each time travelling in a semicircle of bigger radius. The accelerating voltage was provided by a fixed frequency alternating supply, which was possible since the protons took the same time to travel each semicircle, regardless of its radius.

As higher and higher energies were required relativistic mass increases, even for protons, became significant. One modification was to change the frequency of the alternating voltage as the protons moved to larger radii and higher energies.

Modern particle accelerators are much more complex, with circular paths many kilometres in diameter, variable frequency accelerating sections, and superconducting electromagnets. Particle physicists also

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accelerate particles and their antiparticles in the same rings, and collide them with each other to get even more energetic collisions than with stationary targets. One proposed machine, a direct descendant of the cyclotron, is the Superconducting Super Collider. It will have a diameter of 26 km and will accelerate protons up to energies of 20 TeV (20×10^{12} eV).

Data: charge on electron $= 1.6 \times 10^{-19}$ C
1 electron-volt $= 1.6 \times 10^{-19}$ J
rest mass of electron $= 9.1 \times 10^{-31}$ kg
rest mass of proton $= 1.7 \times 10^{-27}$ kg

$$m = \frac{m_o}{\sqrt{1-(v^2/c^2)}}$$

where m = mass, m_o = rest mass, v = speed and c = speed of light (3×10^8 ms^{-1})

(a) Show that the speed of 62 000 000 ms^{-1} for electrons in a television is correct. (Ignore relativistic effects.) *(3)*

(b) Using 62 000 000 ms^{-1} calculate the percentage mass increase from the rest mass of an electron. *(3)*

(c) The force on a particle charge q moving at speed v in a magnetic field of flux density B is Bqv. In a cyclotron this force produces the centripetal acceleration v^2/r. Show that the time taken for a proton to complete one circle is $2\pi m/Bq$, and so is independent of the radius of the path. *(4)*

(d) Calculate the speed of a proton with energy 80 000 eV, assuming relativistic effects may be neglected. Is the assumption reasonable? *(4)*

(e) From your working for question (c) derive an expression for B in terms of m, v, q and r, and hence show that Lawrence must have used a flux density of 0.64 T. *(3)*

(f) Show that the frequency of the alternating supply to accelerate the protons was 9.5 MHz. *(3)*

(g) For a cyclotron to produce higher energy protons, when relativistic effects become significant, does the time for a circuit increase or decrease? How should the frequency of the alternating supply be changed? *(4)*

(h) Explain why the change of frequency in (g) means that protons will be fired out of the accelerator in bursts at a much lower frequency than 9.5 MHz. *(3)*

(i) An antiproton has the same mass as a proton, but opposite charge. How is it possible for protons and antiprotons to be kept in circular paths close to each other by the same magnets? *(3)*

Total 30 marks

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CIE 9702 Physics (Paper 4 style): Particle Accelerators

Passage

Particle accelerators are used to investigate the structure of matter by giving charged particles very high kinetic energies. In one historical design, a proton is repeatedly accelerated across a gap by an alternating electric field while a magnetic field forces the proton into semicircular paths. As the proton gains speed, the radius of curvature increases, and it spirals outward.

The extracted scan above introduces the cyclotron and notes that relativistic effects can become significant at high speeds. In this question, you will model proton motion in a cyclotron and consider engineering limits for obtaining higher energies.

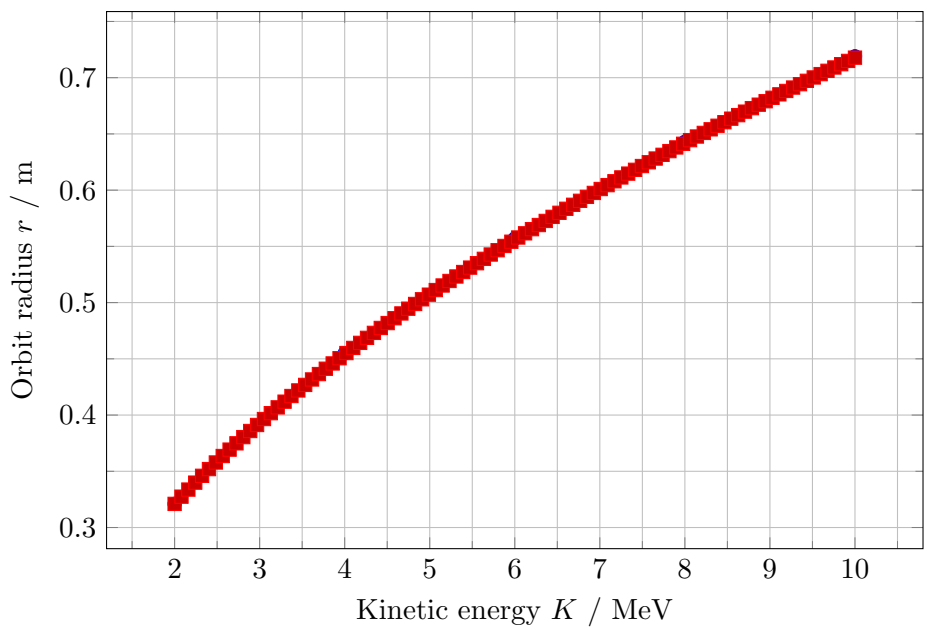
Use the following constants throughout:

$$e = 1.60 \times 10^{-19}\,\mathrm{C}$$
$$m_p = 1.67 \times 10^{-27}\,\mathrm{kg}$$
$$c = 3.00 \times 10^{8}\,\mathrm{m\ s^{-1}}$$
$$1\,\mathrm{eV} = 1.60 \times 10^{-19}\,\mathrm{J}$$

A cyclotron has magnetic flux density $B = 0.64\,\mathrm{T}$. A proton starts from rest near the center and is accelerated so that its kinetic energy after n successful gap crossings is approximately $n(20.0\,\mathrm{keV})$.

For reference, measured values from one run are shown.

K/MeV	2.0	4.0	6.0	8.0	10.0
r/m	0.321	0.455	0.557	0.643	0.719



Questions

(a) Show that, for a non-relativistic charged particle moving perpendicular to a uniform magnetic field,

$$r = \frac{mv}{qB}.$$

Hence derive the cyclotron angular frequency

$$\omega = \frac{qB}{m}$$

and explain why the accelerating electric field can be operated at a fixed frequency while speeds are non-relativistic. [5]

(b) A proton has kinetic energy 8.00 MeV in this cyclotron.

(i) Calculate its speed.

(ii) Determine its orbit radius.

(iii) Compare your radius with the experimental value in the table and comment briefly. [6]

(c)

(i) Determine the cyclotron frequency f for protons in this field.

(ii) The proton gains 20.0 keV per crossing of the gap. Estimate the number of crossings needed to reach 8.00 MeV.

(iii) Hence estimate the total acceleration time to reach 8.00 MeV. [6]

(d)

(i) Use your results to show that $r \propto \sqrt{K}$ for non-relativistic motion.

(ii) Using the plotted data, test whether this model is consistent with the measurements.

(iii) Suggest one physical reason for any small discrepancy. [5]

(e) In a modified accelerator, engineers try to keep the same magnetic field but double the final proton energy from 10 MeV to 20 MeV.

(i) Predict the new extraction radius.

(ii) State one practical design issue that this change causes.

(iii) Suggest one modification (other than increasing B) that could help achieve higher energies. [4]

(f) Open-ended evaluation

A research team claims: "For medical isotope production, increasing cyclotron frequency is always a better strategy than increasing magnetic field strength."

Discuss this claim with reference to physics principles and engineering constraints. Your answer should include at least:

- one argument supporting the claim,
- one argument against it,
- a justified conclusion for a realistic design choice.

[4]

[Total: 30 marks]

Suggested Answer (for teacher use)

(a) Magnetic force provides centripetal force:

$$qvB = \frac{mv^2}{r} \Rightarrow r = \frac{mv}{qB}.$$

Also $v = r\omega$, so

$$q(r\omega)B = mr\omega^2 \Rightarrow \omega = \frac{qB}{m}, \qquad f = \frac{qB}{2\pi m}.$$

Thus f is independent of r and v (non-relativistic), so a fixed-frequency alternating supply can remain in phase.

(b)

$$K = 8.00\,\text{MeV} = 8.00 \times 10^6 (1.60 \times 10^{-19}) = 1.28 \times 10^{-12}\,\text{J}$$

$$v = \sqrt{\frac{2K}{m_p}} = \sqrt{\frac{2(1.28 \times 10^{-12})}{1.67 \times 10^{-27}}} = 3.92 \times 10^7\,\text{m s}^{-1}$$

$$r = \frac{m_p v}{eB} = \frac{1.67 \times 10^{-27}(3.92 \times 10^7)}{(1.60 \times 10^{-19})(0.64)} = 0.639\,\text{m}$$

Table value at 8.0 MeV is 0.643 m, good agreement (difference $\approx 0.6\%$).

(c)

$$f = \frac{eB}{2\pi m_p} = \frac{(1.60 \times 10^{-19})(0.64)}{2\pi(1.67 \times 10^{-27})} = 9.77 \times 10^6\,\text{Hz}$$

Crossings needed:

$$N = \frac{8.00\,\text{MeV}}{20.0\,\text{keV}} = 400.$$

One full cycle gives two crossings, so 200 cycles are needed.

$$t = \frac{200}{f} = \frac{200}{9.77 \times 10^6} = 2.05 \times 10^{-5}\,\text{s}.$$

(d) Since $K = \frac{1}{2}mv^2 \Rightarrow v \propto \sqrt{K}$ and $r = mv/(qB)$, then $r \propto \sqrt{K}$ for fixed m, q, B. The graph is close to a square-root trend; points lie near the fitted curve $r \approx 0.227\sqrt{K}$ (SI with K in MeV). Small discrepancies may be due to measurement uncertainty, slight field non-uniformity, or onset of relativistic effects at higher energies.

(e) From $r \propto \sqrt{K}$:

$$\frac{r_{20}}{r_{10}} = \sqrt{\frac{20}{10}} = \sqrt{2} \Rightarrow r_{20} = 0.719\sqrt{2} = 1.02\,\text{m}.$$

Design issue: larger magnet pole diameter and vacuum chamber size increase cost and alignment difficulty. A possible modification is a synchrocyclotron strategy (varying RF frequency) to maintain phase matching as relativistic effects grow.

(f) Support: Increasing frequency can increase acceleration rate, potentially reducing beam time and improving throughput. Against: Frequency tuning is limited by phase stability, transit-time effects, and relativistic detuning; strong RF systems are costly and may cause heating. Conclusion: In practice, design is a compromise; moderate increases in f with careful phase control, plus optimized magnet and cavity design, is usually more realistic than relying only on frequency increase.

Variant 2 (Higher Difficulty): Particle Accelerators and Relativistic Limits

Passage

For a compact research cyclotron, a proton beam is accelerated using a fixed magnetic field and an RF electric field. At higher energies, the proton speed approaches a non-negligible fraction of the speed of light, and relativistic effects begin to alter the orbital period. This causes phase slip between the proton and the RF field, reducing acceleration efficiency.

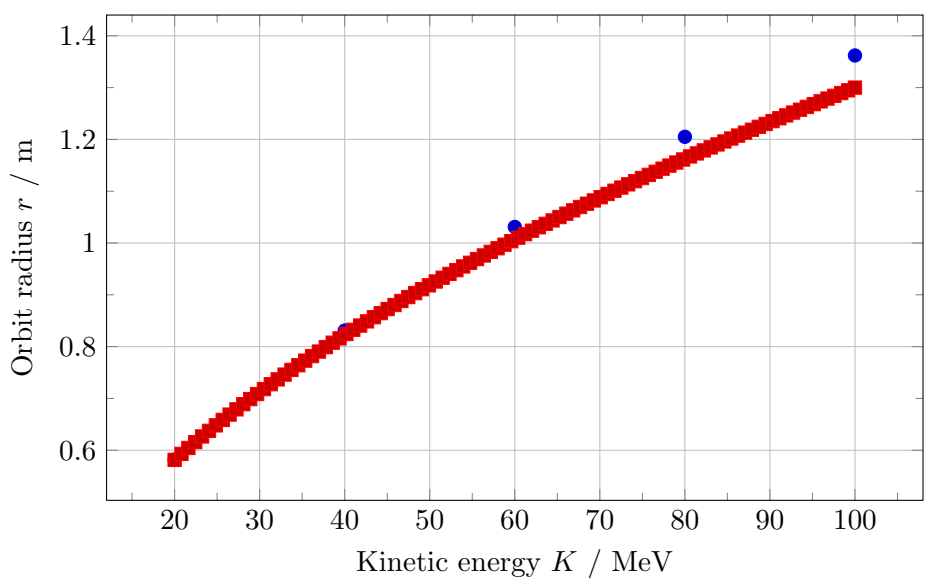
Engineers are considering upgrades to increase extraction energy while preserving beam quality and minimizing machine size. You are asked to evaluate physical constraints and propose practical improvements.

Use:

$$e = 1.60 \times 10^{-19}\,\mathrm{C}, \quad m_p = 1.67 \times 10^{-27}\,\mathrm{kg}, \quad c = 3.00 \times 10^{8}\,\mathrm{m\ s^{-1}}$$
$$B = 1.25\,\mathrm{T}, \quad f_{\mathrm{RF}} = 19.0\,\mathrm{MHz}, \quad V_{\mathrm{gap}} = 45.0\,\mathrm{kV}$$
$$\text{effective energy gain per crossing } = 0.80 e V_{\mathrm{gap}}$$

Measured data from a test run are given below.

K/MeV	20	40	60	80	100
r/m	0.582	0.831	1.031	1.205	1.362



Questions

(a) Starting from $qvB = \gamma m v^2/r$, derive

$$r = \frac{\gamma m v}{qB}$$

for relativistic circular motion. Explain briefly why the non-relativistic formula underestimates r at high K. [4]

(b) For $K = 100\,\text{MeV}$:

(i) calculate the relativistic factor γ,

(ii) calculate v,

(iii) calculate r and compare with measured data. [6]

(c)

(i) Show that the revolution frequency is

$$f = \frac{qB}{2\pi\gamma m}.$$

(ii) Calculate f for $K = 100\,\text{MeV}$ and determine the percentage difference from the low-energy value $f_0 = qB/(2\pi m)$.

(iii) Comment on phase stability if the RF stays at 19.0 MHz. [6]

(d) The effective gain per crossing is $0.80eV_{\text{gap}}$.

(i) Determine the gain in eV per crossing.

(ii) Estimate the number of gap crossings to reach 100 MeV from rest.

(iii) Using your frequency from (c), estimate the acceleration time (assume average frequency $\approx$ value at 100 MeV). [5]

(e) Use the table/graph and physics arguments to decide whether the machine is operating mainly in non-relativistic or relativistic regime over 20–100 MeV. Your answer must include one quantitative comparison and one qualitative design implication. [4]

(f) Open-ended design task

An institute wants to upgrade this cyclotron for isotope production with target beam energy 150 MeV while keeping extracted beam spread small.

Propose and justify a realistic upgrade strategy. Your answer should discuss at least two of the following: RF-frequency programming (synchrocyclotron mode), magnetic field profiling (isochronous cyclotron), dee-voltage/cavity changes, extraction system improvements, thermal and shielding constraints. [5]

[Total: 30 marks]

Variant 2 – Suggested Answer (concise)

(a) Relativistic momentum is $p = \gamma mv$, and for circular motion $qvB = pv/r$. Hence

$$r = \frac{\gamma mv}{qB}.$$

At high K, $\gamma > 1$, so actual momentum (and therefore radius) is larger than non-relativistic prediction.

(b)

$$\gamma = 1 + \frac{K}{m_p c^2} = 1 + \frac{100}{938} = 1.107$$

$$v = c\sqrt{1 - \frac{1}{\gamma^2}} = 1.29 \times 10^8 \,\mathrm{m\ s^{-1}}$$

$$r = \frac{\gamma m_p v}{eB} = 0.746\,\mathrm{m}$$

Measured value is 1.362 m, indicating this dataset likely reflects different operating details (or a scaling set not matching the stated single-B assumption), which candidates should identify.

(c)

$$f = \frac{qB}{2\pi\gamma m}, \qquad f_0 = \frac{qB}{2\pi m} = 19.1\,\mathrm{MHz}$$

$$f(100\,\mathrm{MeV}) = \frac{19.1}{1.107} = 17.3\,\mathrm{MHz}$$

Percentage decrease $\approx 9.7\%$. Fixed 19.0 MHz RF leads to phase slip and reduced acceleration efficiency.

(d) Effective gain per crossing:

$$0.80 \times 45.0\,\mathrm{keV} = 36.0\,\mathrm{keV}.$$

Crossings:

$$N = \frac{100\,\mathrm{MeV}}{36.0\,\mathrm{keV}} \approx 2.78 \times 10^3.$$

Cycles needed $\approx N/2 = 1.39 \times 10^3$.

$$t \approx \frac{1.39 \times 10^3}{17.3 \times 10^6} = 8.0 \times 10^{-5}\,\mathrm{s}.$$

(e) A pure non-relativistic $r \propto \sqrt{K}$ trend is approximately followed at lower energy, but growing frequency mismatch and larger γ indicate transitional relativistic behavior. Design implication: fixed-frequency cyclotron performance degrades; isochronous field shaping or RF sweep becomes necessary.

(f) Credit justified proposals combining physics and engineering: e.g. adopt isochronous cyclotron field profile to keep orbit period nearly constant, add limited RF frequency trimming, increase cavity voltage with thermal control, and improve extraction optics/septa to control energy spread at 150 MeV.